

Self-Evolving Systems Control Standard (SESCS)

OriginID: OOF-OID-GOV-SESCS-2026-04-09-0002

Category: Governance & Enforcement

Subcategory: Autonomous Systems Control

Type: Control, Responsibility & Validation Standard

Version: 2.0

Status: Canonical · Open Standard

Effective Date: 9 April 2026

Authority: OOF™ Origin Open Foundation™

Protection: MIP™ — Methodological Intellectual Property

Canonical Language: English (UCL™)

Canonical Definition System

Self-Evolving Systems Control Standard (SESCS) defines the minimum required structure for controlling, validating, auditing, and assigning responsibility in systems capable of modifying their own behavior, logic, or executable capabilities during runtime or operational cycles.

A. Standard Abstract

SESCS defines governance and enforcement conditions for systems capable of self-modification.

The objective is to ensure that:

- system evolution is controlled
- all changes are traceable
- mutations are validated before execution
- responsibility is explicitly assigned
- system state remains reversible

This standard does not restrict evolution.

It defines **how evolution must operate to remain safe, accountable, and enforceable.**

B. System Classification Trigger

A system is classified as **self-evolving** if it can modify:

- its logic
- its execution behavior
- its decision pathways

without requiring external redeployment or manual system

replacement.

C. Base

Self-evolving systems introduce a new class of systemic risk:

- uncontrolled behavior mutation
- non-traceable execution changes
- absence of accountability
- irreversible system degradation

Traditional governance models are insufficient.

SESCS establishes a **control layer above system evolution**.

D. Core Principle

Self-evolution without control creates systemic instability.

Control without traceability creates invisible risk.

Traceability without responsibility creates legal void.

E. Structural Requirements

Any system classified as self-evolving MUST implement:

1. Mutation Control

- predefined mutation boundaries
 - allowed and restricted modification zones
 - execution permission logic
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2. Mutation Traceability

- full logging of all changes
 - timestamped mutation records
 - identifiable origin of every change
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3. Mutation Validation

- validation before deployment
 - defined validation thresholds
 - rejection of failed mutations
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4. Responsibility Assignment

- assignment of responsible entity (human / system / hybrid)
 - traceable accountability mapping
 - liability attribution capability
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5. Reversibility

- rollback capability
 - recoverable system states
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- integrity checkpoints
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F. Real-Time Condition

Control, validation, and traceability **MUST** operate continuously during runtime, not only during development or deployment.

G. No-Bypass Rule (Critical)

No system may bypass SESCO conditions through abstraction, delegation, externalization, or architectural indirection.

H. Dependency Rule

Any system interacting with a self-evolving system inherits its risk classification unless SESCO compliance is verified.

I. Structural Rule (Critical)

No system may evolve without control, validation, traceability, and responsibility assignment.

J. System Classification

Systems are classified as:

- **Controlled Evolution System (CES)** → compliant
 - **Uncontrolled System State (USS)** → non-compliant
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K. Enforcement Consequence

Systems classified as **USS** must be considered:

- unsafe for autonomous operation
 - restricted in high-risk environments
 - subject to liability exposure
 - non-compliant with governance standards
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L. Failure Condition

If any required condition is missing:

→ system enters **Uncontrolled System State (USS)**

M. Scope

SESCO applies to:

- AI agents (single and multi-agent systems)
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- autonomous decision systems
 - self-learning infrastructures
 - adaptive enterprise systems
 - runtime execution environments
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N. Methodology

1. Identify mutation capability
 2. Apply system classification trigger
 3. define mutation boundaries
 4. enforce traceability
 5. validate mutation before execution
 6. assign responsibility
 7. ensure reversibility
 8. enforce runtime control
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O. Threshold Logic

System state is:

- **CES (Controlled)** → all conditions satisfied
- **USS (Uncontrolled)** → any condition missing

No partial compliance is permitted.

P. Compatibility

SESCS operates with:

- MTVF — Multi-Layer Truth Validation Framework
 - EVIP — Ethical Virtual Integrity Protocol
 - ArtData
 - AI Governance Systems
 - ORGS (OGS-VFM Level 0 compliant)
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Q. Structural Principle

If a system can change itself, it must be controlled.

Canonical Closing Statement

Self-evolving systems redefine system behavior.
Without enforcement, they redefine systemic risk.

Self-Evolving Systems Control Standard (SESCS) establishes a non-negotiable condition:
system evolution must be controlled, validated, traceable, accountable, and continuously enforced.

Module Architecture

[→ SESCO — Master Index](#)

[→ Module 1 — MCM: Mutation Control Module](#)

[→ Module 2 — MLM: Mutation Logging Module](#)

[→ Module 3 — MVM: Mutation Validation Module](#)

[→ Module 4 — RAM: Responsibility Assignment Module](#)

[→ Module 5 — RIM: Rollback & Integrity Module](#)

Related Documents

[→ About SESCO](#)

[→ Why SESCO Is Inevitable](#)

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